Saksham Shrestha

Data Science Student | Python Developer

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TECHNICAL SKILLS

Programming Languages: Python (Pandas, NumPy), SQL, Web Scraping Data Science: Data Analysis, Data Cleaning, Data Visualization Tools: Git, GitHub, Jupyter, VS Code Data Visualization: Matplotlib, Seaborn Databases: SQLite, MySQL, CRUD Operations, Query Optimization

EDUCATION

Bachelor of Science in Computer Science & Information Technology (BSc. CSIT) Amrit Campus, Kathmandu | December 2023 – Present

- Currently pursuing undergraduate degree in Computer Science

Data Science with Python Training Broadway Infosys Nepal, Kathmandu | December 2025 – Present

- Ongoing professional certification in data science and machine learning

NEB 10+2 (Science) Xavier Academy, Kathmandu | 2020 – 2022

- GPA: 3.7 out of 4.0
- Relevant Coursework: Mathematics, Physics

PROFESSIONAL EXPERIENCE

Inventory Associate Buneko Nepal, Kathmandu | July 2023 – August 2023

- Maintained inventory records with 100% accuracy using data management systems
- Performed data entry and verification for stock tracking and management
- Supported operational workflows with attention to detail and accuracy
- Ensured zero deficiencies in inventory logs through systematic verification

PROJECTS

Satellite-Based Crop Health Predictor | Python, XGBoost, Scikit-Learn, Pandas, Streamlit, Planetary Computer API, Joblib

- Ingested multi-modal geospatial and meteorological data by querying Sentinel-2 imagery via the Planetary Computer STAC API and harvesting daily climate metrics from the Open-Meteo API
- Processed high-resolution satellite bands to calculate cloud-masked scene NDVI and engineered 14-day rolling features including moisture deficit, water satisfaction ratio, and heatwave indicators

- Trained and evaluated an XGBoost Regressor on chronologically split time-series data to forecast plant vigor (NDVI) 14 days in advance using MAE and R^2 metrics
- Developed an interactive Streamlit dashboard featuring baseline vs. forecasted NDVI predictions, actual-vs-predicted trend visual charts, and dynamic historical date filters
- Orchestrated an end-to-end modular pipeline automating raw data fetching, feature creation, model execution, and artifact serialization via Joblib for decoupled application serving

Nutritional Parameter Analysis Engine | Python, Scikit-Learn, Pandas, Joblib, Streamlit

- Trained and evaluated classification models (Logistic Regression, Decision Trees, Random Forest) on a 7-feature core nutrient vector to predict food healthiness
- Preprocessed raw data with custom string sanitization, missing-value handling, and normalized numeric inputs using StandardScaler
- Built an interactive Streamlit dashboard featuring dynamic serving-size scaling and traffic light visual status indicators
- Integrated hybrid rule-based health profile guardrails (Diabetic, Low Sodium, Keto) to override ML predictions when critical nutrient limits are breached
- Serialized trained model binaries and scaler artifacts using joblib with an automated heuristic fallback system for robust fault tolerance

Banking Support Agent | Python, LangChain, Gemini API, FAISS, Streamlit

- Built a RAG pipeline that embeds banking FAQ documents into a FAISS vector store using Google Generative AI embeddings
- Implemented semantic similarity search to retrieve the most relevant document chunks for any user query
- Designed a modern LCEL chain with prompt templating to generate grounded, hallucination-free responses via Gemini
- Developed a Streamlit chat interface with session state management for persistent conversation history

F1 Race Day Notifier Bot | Python, Pandas, Viber API, GitHub Actions

- Fetched live Formula 1 session data from the OpenF1 REST API and parsed it into a pandas DataFrame
- Applied datetime filtering logic to detect next-day races and extract location and time details
- Delivered automated race alerts to Viber messenger via the Viber Bot API
- Deployed on GitHub Actions to run on a daily schedule with no manual intervention or local machine required

Stock Market Data Collector & Viewer | Python, MySQL

- Developed Python scripts to fetch and process daily stock market data from APIs
- Performed data cleaning, validation, and normalization to ensure data integrity
- Stored structured financial data in a MySQL database and executed SQL queries for analysis
- Automated CSV exports and implemented a CLI-based search for stock lookup

CERTIFICATIONS

- SQL (Basic) Certification | HackerRank | 2026
- Bronze Honor Recipient | International Youth Math Challenge (IYMC) | 2023
  Ranked in top 15% globally for problem-solving and mathematical reasoning

ACHIEVEMENTS

Xavier Academy Science Exhibition Organizer | 2022

- Successfully organized and coordinated annual science exhibition
- Managed team collaboration and event logistics

Robotics Club Member | Xavier Academy | 2022

- Participated in robotics workshops and team projects
- Developed problem-solving and technical skills through hands-on activities

VOLUNTEER EXPERIENCE

Member | Nepali Pariwar | 2022 – Present

- Raised funds and donated supplies to Heartbeat Foundation Nepal
- Organized weekly community cleaning campaigns for environmental sustainability
- Coordinated donations to local institutions for social welfare

Volunteer | Food for Life Nepal | 2019

- Participated in fundraising campaign for children's welfare and education
- Assisted in event management and community outreach activities

Summary

Computer Science undergraduate with hands-on experience in Python, data analysis, and AI development. Built projects spanning automated API integrations and scheduling systems, data pipelines, and a RAG-based conversational AI agent using LangChain and Google Gemini. Seeking an AI/ML internship to contribute to real-world intelligent systems and deepen expertise in generative and agentic AI.